

SPD token grid

input tensor:
 $B \times T \times F \times R$
 $\times d_0 \times d_0$

Tri-axial SPD Transformer encoder $\times L$

Time axis

Riemannian MHA

4 independent heads

Sequence AddNorm

SPD-FFN

Sequence AddNorm

Frequency axis

Riemannian MHA

4 independent heads

Sequence AddNorm

SPD-FFN

Sequence AddNorm

Region axis

Riemannian MHA

4 independent heads

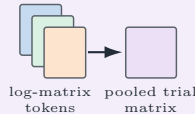
Sequence AddNorm

SPD-FFN

Sequence AddNorm

one encoder layer in the reported configuration

Log map & weighted pooling



Log-Euclidean MDM

learned class
prototypes
distance-based
decision

Class
prediction